
Position: Human-Centric Vision Requires Topological Generalization Beyond Fixed Skeletal Topologies

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Abstract

In this position paper, we argue that human-centric vision requires skeletal-topology generalization beyond fixed skeletons. Mainstream pose and body pipelines enforce a fixed skeleton graph with an indexed joint list and fixed adjacency, so the fixed joint inventory does not cover structural absence and anatomical absence becomes an ill-posed target for individuals with limb deficiencies. Anatomical absence is not a visibility state, so masking and forced completion can hide structural mismatch and produce hallucinated structure that contaminates downstream reasoning in prosthesis-facing settings. We argue that scaling data and model size alone does not resolve this mismatch while the skeleton schema remains fixed, and this is not a niche concern because these failures affect a large population and reach accessibility-facing systems. We advocate instance-adaptive skeletal topology, where a model estimates anatomical existence and exposes valid connectivity, either by predicting edges directly or deriving them from explicit structural rules, to produce an instance-specific skeleton graph that supports consistent inference and evaluation. We outline measurement upgrades, including existence-aware annotations with explicit absence semantics, skeletal-topology-aware scoring, and hallucination-under-absence penalties, and we close with a call to action for dataset curators, benchmark organizers, and model builders to treat morphological variation as a first-class generalization axis.

1. Introduction

Human-centric computer vision functions as an interface between people and digital systems, spanning assistive use, wearables, telepresence, and embodied AI. Mainstream pose and body pipelines represent the human through a *fixed skeletal topology*, namely a predefined skeleton graph with an indexed joint inventory and fixed adjacency, often paired with parametric body models that assume fixed mesh topology. This choice stabilizes annotation, tooling, and evaluation, so COCO-style keypoints and SMPL-style models become shared interfaces that many methods and benchmarks reuse and optimize against (Lin et al., 2014; Loper et al., 2015; Jin et al., 2020; Cao et al., 2021; Sun et al., 2019; Xu et al., 2022).

The same interface also defines a boundary for who benefits from claimed “generalization,” and Figure 1 previews the resulting mismatch. For individuals with limb deficiencies, the fixed joint inventory does not cover structural absence, so fixed skeletal topology shifts from a helpful prior to a mis-specified schema that is incomplete for structural variation. Anatomical absence is not a visibility state (Jin-s13 & COCO-WholeBody contributors). When a model must emit coordinates or tokens for non-existent joints, a fixed-topology objective encourages template completion with hallucinated structure, or masking hides mismatch as silent error and unstable uncertainty. This mismatch propagates to downstream stages that consume skeleton structure, including 3D fitting, tracking, pose-conditioned understanding, rehabilitation and clinical motion analysis, and interaction pipelines where structural semantics constrain feasible inference. Template-locked body models amplify the issue because fixed surface and joint topology forces optimization within a canonical parameterization that does not align with limb-deficient morphology (Loper et al., 2015; Mahmood et al., 2021).

This is not a niche concern. Individuals with limb deficiencies form a large population worldwide (World Health Organization, 2022). WHO also estimates that more than 2.5 billion people need one or more assistive products, which shows how interface-layer modeling gaps reach accessibility-facing systems at population scale (World Health Organization, 2024). If a system collapses non-

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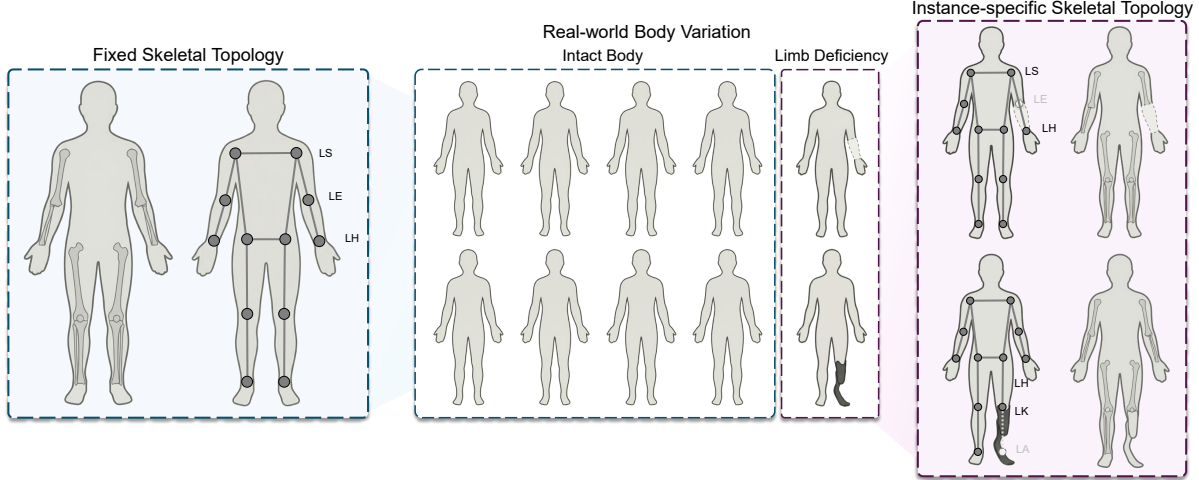


Figure 1. A schematic of fixed-skeletal-topology assumptions, limb-deficiency variation, and instance-adaptive skeletal topology. **Left:** an intact-body interface that treats the joint inventory and adjacency as constants. **Middle:** a population-scale reminder that limb deficiency is common, so fixed-schema assumptions shape who benefits from claimed generalization. **Right:** two representative cases with distinct structural changes, where intercalary deficiency rewires adjacency while terminal amputation removes distal joints, motivating instance-adaptive prediction of joint existence and connectivity.

existence into a canonical index, or invents missing structure, it does not represent anatomy as it is. A fixed-skeletal-topology pipeline then optimizes for template completion rather than topology-consistent body structure, and the topological notion of boundary formalizes why absence is a semantic contrast rather than merely weak evidence (Stanford Encyclopedia of Philosophy, 2024).

Position: Human-centric pose and body pipelines require skeletal-topology generalization beyond fixed skeletons, treating skeletal topology as an input-conditioned object to be inferred rather than a constant schema to be fit. We do not argue to abandon skeletal topology. We argue to stop hard-coding a single canonical skeletal topology as the only valid output schema. Instead, we advocate *instance-adaptive skeletal topology*, where a model jointly predicts joint existence and skeletal connectivity, producing an instance-specific skeleton graph that aligns supervision, inference, and evaluation with anatomical presence.

This position remains technically plausible because machine learning infers relational structure rather than assuming it, including interaction-graph prediction and discrete structure learning (Kipf et al., 2018; Franceschi et al., 2019). Graph and attention mechanisms also make explicit that connectivity governs information flow, so connectivity choices under masking shape what evidence reaches what outputs (Velickovic et al., 2018). Skeletal-topology generalization therefore makes “understanding the human body” measurable as structure inference rather than template filling.

We do not claim that this position paper completes the empirical validation of every failure mode. Our claim is a falsifiable representation-level claim: if anatomical non-existence

is collapsed into visibility-style labels and fixed adjacency, then supervision, inference, and evaluation can become insensitive to structurally invalid predictions. This claim can be tested by absence-explicit annotations, topology-aware metrics, and ablations comparing fixed topology, fixed topology with masking, existence-only heads, and full instance-adaptive topology.

2. Preliminaries

2.1. Scope

This paper studies *skeletal-topology and semantics mismatch* in human-centric pose and body pipelines that represent the body as a fixed skeleton graph. A fixed joint inventory does not cover structural absence, so the output schema is incomplete for structural variation in individuals with limb deficiencies. We use *skeletal topology* to denote the skeleton-graph specification, including (i) an *indexed joint inventory* and (ii) a *fixed adjacency* that encodes anatomical or kinematic dependencies and governs information flow, constraints, and downstream reasoning. Our focus is not the output parameterization (coordinates, heatmaps, or tokens), but the assumption that joint inventory and adjacency are constants shared by all people, as in COCO-style keypoint schemas and many SMPL-family pipelines (Lin et al., 2014; Jin et al., 2020; Loper et al., 2015; Pavlakos et al., 2019).

We restrict attention to *structural* questions that persist as data scale and backbone capacity increase. Specifically, morphology can change the appropriate joint inventory and adjacency, so the target structure becomes input-conditioned. Individuals with limb deficiencies provide a direct stress

test because limb absence creates cases where canonical downstream joints are anatomically non-existent rather than merely unobserved (Ying et al., 2025; Du et al., 2026).

Our scope is the body-interface layer: the semantic schema through which downstream systems consume pose, mesh, or skeleton-like structure. We do not claim that every disability-related perception problem reduces to skeletal topology, nor that a single architecture solves all cases. We focus on cases where the canonical node inventory or adjacency is anatomically invalid.

2.2. Terminology and semantics

We separate three concepts that practice often conflates.

Occlusion. A part is *occluded* when it exists but is not visible due to viewpoint, clothing, self-occlusion, or objects. Occlusion weakens evidence but does not change anatomy, so the joint remains part of the joint inventory.

Anatomical absence. A canonical anatomical type is non-existent for an instance when the individual does not have that anatomical entity as part of their body structure; its coordinate target is therefore undefined, not merely unobserved. Absence changes anatomy, so canonical downstream joints on the missing limb become *non-existent* targets whose coordinates are semantically undefined under a fixed skeleton schema (Ying et al., 2025; 2026; Du et al., 2026).

Hallucination under absence. We use *hallucination under absence* for confident predictions of structure that lack evidential and semantic support under the individual’s morphology. This includes high-confidence distal joints on an absent limb, or adjacency that implicitly assumes non-existent nodes.

Annotation decision rule. Absence labels are semantic labels, not visibility labels. A type is labeled non-existent only when image evidence, dataset context, or adjudication supports anatomical non-existence for the depicted individual. If the part may exist but is not visible, the state is present-occluded; if evidence is insufficient, the instance should be adjudicated or excluded from absence-specific scoring. The purpose is not to assume perfect single-frame certainty, but to prevent “not visible,” “not labeled,” and “non-existent” from collapsing into the same training target.

2.3. Problem abstraction: structured prediction under schema mismatch

Let x denote an input (image or video) depicting a person, and let $T(x)$ denote the person-specific skeletal topology consistent with anatomical presence in x . A fixed-skeletal-topology pipeline assumes a canonical schema $S_0 = (J_0, A_0)$ and learns

$$f_\theta : x \mapsto \hat{y}_{S_0},$$

where $\hat{y}_{S_0}$ is indexed by J_0 and constrained by A_0 . Here, J_0 denotes the canonical joint or node-type inventory, and A_0 denotes the canonical adjacency among those types. When $T(x) \neq S_0$, the output space can include joints and relations whose semantics are undefined under anatomical absence, yielding *schema mismatch*. $T(x)$ can be represented as a typed body graph. Learning $T(x)$ does not require a specific architecture: a CNN, transformer, or foundation backbone may feed a head that predicts type-level existence and either predicts edges directly or applies a published structural rule.

Skeletal-topology generalization instead treats topology as input-conditioned and predicts it explicitly. An *instance-adaptive skeletal topology* approach aims to model

$$f_\theta : x \mapsto (\hat{J}(x), \hat{A}(x), \hat{y}_{\hat{J}(x), \hat{A}(x)}),$$

where $\hat{J}(x)$ is a variable-length joint inventory with per-joint existence and type labels, and $\hat{A}(x)$ is an instance-specific adjacency (or an equivalent conditional routing rule).

2.4. Assumptions and what the paper does not claim

We state the following assumptions to clarify the debate in Section 4.

Skeletal topology is not optional. We do not argue for unstructured outputs. Connectivity shapes information flow and constraints, so masking a fixed schema does not preserve semantics when adjacency stays fixed (Velickovic et al., 2018).

More data is not a sufficient answer to a mis-specified output schema. Our critique targets cases where additional data does not resolve ambiguity because the fixed schema forces non-existent joints into canonical output slots, leaving targets undefined under absence (Ying et al., 2025; 2026).

We target the skeleton layer, not a specific 3D body family. We use SMPL-family pipelines as representative template-locked systems, but the position concerns the skeleton-graph interface and its role in shared supervision, inference, and downstream reasoning (Loper et al., 2015; Mahmood et al., 2021).

Absence labels need not be perfectly decidable from every single image. Our claim is not that all images contain enough evidence to resolve anatomy. It is that dataset semantics and model outputs should preserve the distinction when the benchmark or adjudication supports it, rather than forcing all uncertainty into visibility-style missingness.

3. Fixed Skeletal Topology as Mis-specification

This section makes our position testable by stating a *mechanism chain*, a minimal causal account that connects a fixed skeletal-topology interface to three measurable failure

modes and their downstream consequences. Our claim is not that fixed skeletal topology fails only in a narrow corner case. Instead, we argue that a *fixed interface* that hard-codes an indexed joint inventory and a fixed adjacency can make the joint inventory unable to cover structural absence, can route evidence along anatomically invalid dependencies, and can leave evaluation insensitive to decision-relevant structural errors. Figure 2 summarizes the end-to-end chain that motivates this section, and Sections 3.1–3.4 unpack each link. We use individuals with limb deficiencies as a structural counterexample and a high-risk stakeholder population, because limb deficiency makes *node non-existence* and *adjacency change* factual and is not well explained as mere occlusion (Ying et al., 2025).

3.1. Fixed indices create semantically undefined targets under anatomical absence

This subsection corresponds to the *Schema* and *Labels* links in Figure 2. A fixed-skeletal-topology pipeline defines a canonical keypoint schema with K indexed joints, such as COCO-17 (Lin et al., 2014). Prediction is then forced into K output slots, implemented as coordinate channels, heatmaps, or tokens, and each slot is assumed to denote a stable anatomical entity. Under anatomical absence, this assumption breaks. If a joint does not exist as an anatomical entity, its canonical slot does not denote an uncertain variable. Instead, the slot becomes a semantically undefined target under the fixed schema.

A common response reuses COCO-style visibility flags $v \in \{0, 1, 2\}$ as a proxy for absence. This does not resolve the semantics mismatch because v encodes annotation availability and visibility, not anatomical existence (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020). Two consequences are immediate.

Consequence 1: supervision becomes a dataset convention rather than an existence-consistent state. When $v = 0$ is used, the keypoint is treated as not labeled and is typically removed from the loss or heavily down-weighted (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020). The model therefore learns when supervision disappears under a dataset format, instead of learning an explicit state that represents non-existence as a first-class outcome.

Consequence 2: the proxy does not communicate existence semantics to downstream modules. Because v is not an anatomical claim, passing $v = 0$ downstream does not state that a joint is non-existent. It states that the joint annotation is unavailable under the dataset format (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020). Downstream modules that require morphology-consistent structure face a choice between assuming the canonical joint exists, inventing it, or suppressing it with heuristics, indicating

that the output schema does not encode absence semantics consistently under morphology (Ying et al., 2025).

3.2. Fixed adjacency distorts information flow under morphology change

This subsection corresponds to the *Schema to Prediction* link in Figure 2. Fixed skeletal topology also commits to a canonical adjacency. This matters because modern pipelines rely on relational inductive bias, via message passing on skeleton graphs, attention over structured tokens, or kinematic constraints that assume fixed chains (Velickovic et al., 2018). When morphology changes, adjacency is not an implementation detail. It defines which evidence can influence which predictions.

A common workaround keeps the canonical adjacency and masks joints that are treated as missing. Masking under fixed adjacency does not preserve semantics. A masked joint either blocks a path or becomes a degenerate relay, and both cases shift feature propagation to intact parts. This is especially problematic when limb deficiency introduces functionally meaningful endpoints and compensatory dependencies that do not reduce to the same chain with blanks (Ying et al., 2025). As a result, fixed adjacency can bias representation learning toward template-consistent completion even when the input supports a different structure.

This argument does not require every edge to be learned from scratch. In many cases, connectivity can be deterministically derived from predicted node types and existence states. The requirement is that the final output expose an instance-specific valid topology and that edge validity be scoreable. Intercalary deficiencies, residual endpoints, and body-device contact anchors expose the limitation of a fixed adjacency because the valid graph is not always the intact graph with distal nodes blanked out.

3.3. OKS-style measurement can create metric blind spots under non-existence

This subsection corresponds to the *Metrics* link in Figure 2. Even if a model encodes uncertainty internally, evaluation can still hide the failure mode that matters under anatomical absence. In COCO-style keypoint scoring, OKS and AP include only labeled ground-truth keypoints with $v > 0$, so unlabeled keypoints do not contribute to the headline score (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors; Jin et al., 2020). Standard evaluation implementations can also ignore ground-truth instances that contain zero labeled keypoints in the keypoint task, which further weakens penalties for confident structure in regions where targets are unlabeled (cocodataset & COCO API contributors). As a result, a method can output high-confidence structure in anatomically absent regions and still avoid proportional penalty in the headline metric, even when

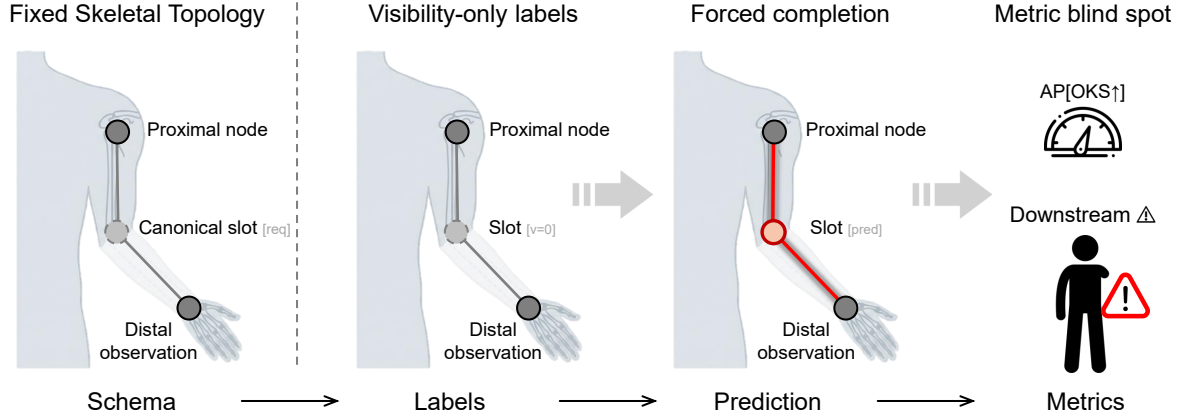


Figure 2. Mechanism chain from fixed skeletal schemas to metric blind spots. This schematic illustrates a fixed-schema slot-level failure rather than a literal anatomical case: the intermediate node denotes a canonical joint slot expected by the fixed topology, while distal visual evidence may correspond to a residual endpoint, prosthetic/device structure, or other observation outside the canonical body schema. The chain shows how non-existence can be represented only through visibility-style conventions, how template-consistent completion can arise, and how OKS-based scoring can under-measure structural errors that matter for downstream use.

the structural error is decision-relevant.

A minimal example. Consider a ground-truth instance where distal joints on an absent limb are encoded with $v = 0$ under the dataset format. In the standard OKS computation, only keypoints with $v > 0$ enter the OKS average, so predictions on these absent-joint slots can have no effect on OKS unless a separate absence-aware penalty is added (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors). If the instance has `num_keypoints = 0`, standard evaluation can mark it as ignored for keypoint evaluation, so it can contribute little to no penalty even when predictions place confident joints in the absent region (coco-dataset & COCO API contributors).

This sensitivity interacts with diagnosis effects. Measured pose performance depends on attributes including visible keypoint count and scene clutter (Ronchi & Perona, 2017). In our setting, this is not a minor nuance. It indicates that missingness interacts with measurement, not only with inference, which makes data-sufficiency arguments incomplete when the output schema does not encode absence semantics consistently. Related work explores evaluation signals that do not rely on ground-truth keypoints, highlighting the need to make evaluation assumptions explicit (Wang et al., 2025).

3.4. Schema mismatch propagates into system-level distortion

This subsection corresponds to the *Prediction to Downstream* consequence in Figure 2. Skeletal predictions act as an interface for downstream modules, including 3D fitting, tracking, pose-conditioned activity understanding, rehabilitation and clinical motion analysis, and human-device interaction (Bogo et al., 2016; Andriluka et al., 2017; Yan et al.,

2018; Stenum et al., 2024; Farulla et al., 2016). When the interface encodes a fixed joint inventory and fixed adjacency, downstream stages inherit the mismatch even if a headline keypoint score appears acceptable.

Template-locked body models amplify this issue because fixed surface and joint topology constrains optimization to explain observations within a canonical parameterization (Loper et al., 2015; Pavlakos et al., 2019; Mahmood et al., 2021). Under anatomical absence, this can force invented structure or can distort geometry to satisfy constraints that presume non-existent joints and chains, yielding failures that are structural rather than cosmetic.

The required remedy differs by downstream use. In 3D fitting, topology mismatch should trigger topology-consistent fitting or abstention rather than template-forced limb completion. In tracking, node existence should be persistent over time so absent joints do not become unstable trajectories. In clinical and rehabilitation analysis, trajectory definitions should condition on valid endpoints and present joints rather than assuming the intact chain. In prosthesis-facing and human-device interaction systems, benchmarks should state whether visible structure is body, device, contact anchor, or out of body schema.

4. Alternative Views

A position remains scientific only if it addresses credible alternatives and stays falsifiable under decisive evidence (International Conference on Machine Learning, 2026). We therefore state four reasonable opposing views and the single evidence condition that changes our mind in each case.

Table 1. Debate map for Alternative Views on topology generalization beyond fixed skeletons. Decisive evidence is assessed under a shared absence-explicit dataset package and topology-aware evaluation (Gebru et al., 2018; Mitchell et al., 2018; AI, 2023).

Alternative view (one-line)	Decisive evidence (what changes our mind)	Who acts (deliverables)
A: More data + limb-def state. Scale limb-def data and add a dedicated state inside fixed indexed slots.	Fixed-schema + state matches instance-adaptive topology on existence and edge correctness across limb strata, with low hallucination under absence-explicit targets.	Benchmarks: absence targets, topology-aware eval. Models: calibrated existence outputs.
B: Superset + masking. Keep fixed indices and adjacency, mask non-existent joints in training and inference.	Masking-based fixed-topology matches instance-adaptive topology on edge correctness under connectivity stress tests, without reliance on canonical completion behavior.	Benchmarks: connectivity stress suite. Models: routing ablations.
C: Existence heads + small extension. Add existence heads and a small endpoint extension, keep global adjacency fixed.	With fixed adjacency, existence-aware systems match instance-adaptive topology on edge correctness and endpoint behavior, with comparable hallucination under absence-explicit targets.	Datasets: endpoint types, mappings. Models: edge policy reporting.
D: Practicality & reliability. Adaptive topology adds variance and risk, fixed schemas remain the safer scientific choice.	Under shared tooling and shared topology-aware metrics, fixed schemas show lower topology error and lower hallucination, and lower variance across independent implementations.	Benchmarks: reference code, canonicalization. Reviewers: topology-aware reporting.

4.1. Negative A: More data plus a limb-deficiency state makes fixed schemas sufficient

Opposing view. Fixed skeletal schemas remain sufficient if we (i) scale limb-deficiency data and (ii) add a limb-deficiency state inside existing indexed joint slots, so the model outputs a dedicated “deficient” label instead of hallucinated coordinates.

Response. This still treats topology change as a label on a fixed index set and keeps the hypothesis class template-locked. It does not represent cases that require adjacency rewiring or endpoint landmarks outside the canonical joint inventory (Ying et al., 2025). COCO-style keypoint flags reflect labeling and visibility conventions rather than anatomical existence, so mapping “deficient” into suppression or ignore pathways risks learning dataset bookkeeping instead of morphology-consistent semantics (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors; Jin et al., 2020). We change our mind only if a fixed-schema method with this state achieves low hallucination under absence-aware targets and matches topology-aware correctness, including edge correctness, under limb-level stratification (Ying et al., 2025).

4.2. Negative B: Canonical superset with masking preserves correctness

Opposing view. A canonical skeleton acts as a superset, so we keep fixed indices and adjacency and mask absent joints during training and inference, preserving tooling and comparability while avoiding penalties on non-existent joints.

Response. Masking changes which slots contribute to loss but does not change the fixed index-locked representation that the model learns, and fixed adjacency still de-

fines the routing scaffold that governs information flow (Velickovic et al., 2018). Intercalary cases require adjacency changes rather than blanking distal nodes, and limb-deficiency-relevant endpoints are not representable as “missing canonical joints” (Ying et al., 2025). We change our mind only if masking-based fixed-topology methods match instance-adaptive methods on topology-aware metrics, including edge correctness and hallucination under absence, without relying on canonical completion behavior under stratified stress tests (Ying et al., 2025).

4.3. Negative C: Existence heads plus small schema extension is enough

Opposing view. We keep the fixed skeletal schema, add existence-aware heads that separate occlusion from non-existence, and allow a minimal extension for limb-deficiency landmarks such as residual endpoints, while keeping global adjacency unchanged.

Response. This strengthens supervision, but it does not deliver topological generalization unless connectivity becomes input-conditioned. With fixed adjacency, message passing and constraints still operate on an anatomically invalid scaffold in cases that require rewiring, so existence prediction remains an add-on rather than a structural representation (Velickovic et al., 2018). Small node-type extensions help, but they do not ensure consistent graph semantics when the backbone routes information through a canonical chain (Ying et al., 2025). We change our mind only if existence-aware fixed-connectivity systems recover the same functional structure as instance-adaptive topology under topology-aware scoring, including edge correctness and endpoint behavior, with comparable hallucination rates (Ying et al., 2025).

4.4. Negative D: Practicality and reliability favor fixed schemas

Opposing view. Instance-adaptive skeletal topology adds failure modes, increases annotation and evaluation burden, and reduces reproducibility, so the safer choice is to keep fixed schemas and improve them incrementally.

Response. Fixed skeletal schemas provide stable interfaces and shared infrastructure, but these advantages do not guarantee semantic validity under morphology change. If absence collapses into unlabeled fields or proxy visibility, evaluation can miss decision-relevant structural error and yield reliable scores that remain structurally wrong (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors; Jin et al., 2020). Instance-adaptive topology increases engineering risk, but it makes existence and connectivity failures explicit and supports ablations that separate existence prediction risk from connectivity prediction risk (Ying et al., 2025). We change our mind only if, under a shared absence-explicit dataset package and shared topology-aware evaluation, fixed-schema approaches show lower topology error and lower hallucination while also exhibiting lower variance across independent implementations.

5. Blueprint: Instance-Adaptive Skeletal Topology

This section specifies a minimal *output contract* plus a minimal *measurement contract*, so the position remains testable without committing to a specific architecture (Geburu et al., 2018; Mitchell et al., 2018; AI, 2023). The contract keeps skeleton structure as an inductive bias, while making *node existence*, *node types*, and *connectivity* input-conditioned prediction targets (International Conference on Machine Learning, 2026).

Inputs. Let x denote an image or a video clip depicting a person.

Outputs. A system predicts an instance-specific graph and any associated per-node outputs:

$$f_{\theta}(x) \mapsto (\hat{\mathcal{V}}(x), \hat{\mathcal{E}}(x), \hat{y}(x)), \quad (1)$$

where $\hat{\mathcal{V}}(x)$ is a variable-size node set with typed existence, $\hat{\mathcal{E}}(x)$ is an instance-specific edge set (or an equivalent routing rule), and $\hat{y}(x)$ denotes task outputs defined on $\hat{\mathcal{V}}(x)$ and $\hat{\mathcal{E}}(x)$.

This contract is architecture-agnostic. The visual backbone may be a CNN, transformer, pose foundation model, or multimodal foundation model. What changes is the body-interface head: it must expose type-level existence and an instance-specific edge policy. The edge policy may be directly predicted or deterministically derived from predicted

node types and existence states, as long as the resulting topology is explicit and evaluable.

Node set with typed existence. Nodes represent anatomically present and task-relevant landmarks:

$$\hat{\mathcal{V}}(x) = \{(\hat{t}_i, \hat{p}_i, \hat{\ell}_i)\}_{i=1}^{\hat{K}(x)}, \quad \hat{p}_i \in [0, 1]. \quad (2)$$

Here $\hat{t}_i$ is a node type from a published vocabulary, $\hat{p}_i$ is an existence probability, and $\hat{\ell}_i$ is a location or region support. Existence is a semantic claim about anatomical presence rather than a proxy for visibility, so *absence* does not collapse into *occlusion* or unlabeled fields (Ying et al., 2025).

Input-conditioned connectivity. Connectivity is made explicit rather than globally assumed:

$$\hat{\mathcal{E}}(x) \subseteq \hat{\mathcal{V}}(x) \times \hat{\mathcal{V}}(x). \quad (3)$$

The system either predicts an edge set or derives it from a published type-conditional rule. In both cases, the resulting topology must be instance-specific, explicit, and evaluable. This matters because connectivity governs information flow and constraints in graph and attention formulations (Velickovic et al., 2018).

Canonicalization for interoperability. To preserve comparability under variable $\hat{K}(x)$, a method publishes (i) a type vocabulary, (ii) deterministic canonicalization rules for nodes and edges, and (iii) a stable encoding that maps $(\hat{\mathcal{V}}(x), \hat{\mathcal{E}}(x))$ into a shared evaluation space (Mitchell et al., 2018).

Downstream interface behavior. If a downstream module assumes a fixed template, the method specifies an explicit policy for topology mismatch, including abstention or conditional selection of topology-consistent representations, instead of forcing template completion (Mahmood et al., 2021; AI, 2023).

5.1. Minimal measurement protocol

This subsection defines the minimum evidence needed to evaluate the scientific claim, namely that *absence* and *topology rewrite* become scoreable structural variables rather than unlabeled exceptions.

Data semantics. The dataset distinguishes *present-visible*, *present-occluded*, and *non-existent* for each canonical type, since COCO-style $v \in \{0, 1, 2\}$ marks label availability and visibility rather than anatomical existence (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020). At least one endpoint-type landmark is included for limb-deficiency cases, so functionally salient structure does not remain out of schema (Ying et al., 2025). A ground-truth connectivity rule or explicit typed edges are provided, plus minimal strata (limb level, laterality, and embodiment state when applicable) (Ying et al., 2025).

Metrics. Let τ denote a canonical node type with ground-truth state $s(\tau) \in \{\text{vis}, \text{occ}, \text{abs}\}$, abbreviating present-visible, present-occluded, and non-existent. After canonicalization, define the type-level existence score $\hat{p}(\tau) = \max_{i: \hat{t}_i = \tau} \hat{p}_i$, with $\hat{p}(\tau) = 0$ if no predicted node has type τ . A paper reports: (i) **node existence F1**, by thresholding $\hat{p}(\tau)$ against the binary target $\mathbf{1}[s(\tau) \neq \text{abs}]$ and aggregating per type and per slice; (ii) **edge F1**, by computing precision, recall, and F1 between predicted and ground-truth typed edge sets after canonicalization; and (iii) **hallucination under absence**, e.g., $\text{FPR}_{\text{abs}}(\delta) = |\mathcal{A}|^{-1} \sum_{\tau \in \mathcal{A}} \mathbf{1}[\hat{p}(\tau) \geq \delta]$, where $\mathcal{A} = \{\tau : s(\tau) = \text{abs}\}$, reported with threshold sensitivity or absent-region confidence mass for heatmap outputs. If COCO-style scores are also reported, the paper states how unlabeled joints and ignored instances affect the standard metric signal (coco-dataset & COCO API contributors).

Stress tests and reporting. A minimal suite includes an **absence versus occlusion control**, a **connectivity sensitivity** test that isolates routing effects under masking or fixed adjacency, and stratified slices over limb-level and laterality. Each paper reports the type vocabulary, canonicalization rules, metric thresholds, and at least one downstream feasibility proxy that depends on topology rather than only coordinate error (Geburu et al., 2018; Mitchell et al., 2018; AI, 2023). A minimal ablation suite should compare fixed topology, fixed topology with masking, fixed node set with an existence head, fixed node set with existence and endpoint extension but no edge adaptation, and full instance-adaptive topology.

6. Discussion

Human-centric pose and body pipelines often treat a fixed skeletal topology schema as the shared coordination layer for supervision, inference, and evaluation. Our position reframes this design choice as a representation and measurement limitation, not as a corner-case data gap. Individuals with limb deficiencies provide a structural counterexample because two facts remain unavoidable: some canonical nodes are non-existent targets, and some canonical adjacencies are anatomically wrong, so a fixed joint inventory does not cover structural absence and the output schema is incomplete for structural variation under morphology change (Ying et al., 2025).

A representation and measurement boundary. Fixed-index pose formats often encode missingness through label availability or visibility rather than anatomical existence, so *absence* collapses into “unlabeled” fields instead of becoming a first-class semantic state (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020). Standard evaluation further weakens the feedback loop because ignored or unlabeled keypoints reduce penalties for structure that a model predicts in regions where joints do not exist (coco-

dataset & COCO API contributors). Under this boundary, a system can optimize template completion while preserving competitive aggregate scores, even when structural correctness fails for people who violate the canonical schema. Section 5 responds by making node existence, node types, and connectivity explicit prediction targets, and by requiring canonicalization rules that keep comparisons reproducible when output structure varies.

Human-facing implications: reliability and representation. Body representations sit upstream of accessibility-facing systems, clinical motion analysis, and human-device interaction. When a representation forces a person into a canonical template, hallucinated limbs and topology-mismatched kinematic chains create reliability risks for downstream components that assume consistent structure. They also create representational risks because the system implicitly treats an able-bodied template as the default identity representation. This matters at population scale because assistive needs are large and persistent (World Health Organization, 2024). Skeletal-topology generalization supports accountability because it exposes the structural object that downstream reasoning consumes, which enables audits that aggregate scores can hide (Geburu et al., 2018; Mitchell et al., 2018; AI, 2023).

ML implications: understanding as structured generalization. From an ML perspective, limb deficiency is not only a fairness setting. It tests whether a model represents anatomy as an input-conditioned structure or as a fixed template that it completes when evidence is weak (Ying et al., 2025). Related patterns appear whenever the correct relational schema depends on the instance, including articulated objects with missing components, modular tools, and human-device assemblies. The core question is whether semantics remain stable when structure varies, and whether evaluation remains sensitive to the resulting errors.

Scope, limitations, and open questions. This paper does not prescribe a single algorithmic solution. It specifies a minimal output contract that makes claims falsifiable by separating absence from occlusion, scoring structural correctness, and penalizing hallucination under absence (Ying et al., 2025). Open questions include interoperable type vocabularies across tasks, canonicalization that preserves meaningful variation, and uncertainty reporting that keeps absence and occlusion separable in downstream use (AI, 2023).

Practicality under scarce data. Current benchmark scarcity should be treated as a dataset-design limitation, not as evidence that affected users are out of scope. Strong pretrained visual, pose, or multimodal backbones can still be used; the requirement is that the final body interface expose existence and topology rather than projecting all outputs into a fixed intact-body schema. Simulation may support

controlled ablations and targeted stress tests, but it should remain a supplement rather than a replacement until transfer to real prosthesis-facing, rehabilitation, or accessibility settings is validated.

7. Call to Action

Human-centric vision often routes supervision, inference, and evaluation through a fixed skeletal output schema. When absence collapses into “unlabeled” or “not visible”, systems can score well while emitting structurally invalid bodies (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors; Jin et al., 2020). A practical response is to make absence semantics and topology errors explicit, then assign concrete responsibilities (International Conference on Machine Learning, 2026; Gebru et al., 2018; Mitchell et al., 2018; AI, 2023).

Dataset designers.

- **Encode absence semantics.** Tri-state per canonical type: present-visible, present-occluded, non-existent, with boundary-case examples (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020; Gebru et al., 2018).
- **Add functional types.** Residual endpoints and (when relevant) device-contact anchors, with a published type vocabulary (Ying et al., 2025).
- **Expose topology ground truth.** Typed edges or a deterministic adjacency rule, plus minimal slices (limb level, laterality, embodiment state) (Ying et al., 2025; Gebru et al., 2018).

Benchmark organizers.

- **Score structure.** Node existence F1 (non-existent as a class) and edge F1 in a canonicalized typed space.
- **Score hallucination.** Absent-type false positive rate and absent-region confidence summary, without treating non-existence as “unlabeled” (cocodataset & COCO API contributors).
- **Ship reference evaluation.** Canonicalization utilities, scoring scripts, and a reporting template that includes an absence-versus-occlusion control and a connectivity sensitivity test (Ying et al., 2025; Gebru et al., 2018; Mitchell et al., 2018; AI, 2023).

Model builders.

- **Predict topology, not blanks.** Typed existence plus input-conditioned connectivity, rather than fixed-index template completion (Ying et al., 2025).
- **Document semantic contracts.** Absence versus occlusion definitions, confidence calibration protocol, and a mismatch policy for template-locked downstream stages (Mahmood et al., 2021; Mitchell et al., 2018; AI, 2023).
- **Prove the causal link.** Ablations that isolate (i) existence modeling, (ii) edge adaptation, and (iii) type extension

effects.

Tooling maintainers and deployers.

- **Make topology-aware semantics deployable.** First-class non-existent fields in schemas and loaders, drop-in topology metrics, and deployment gates that require topology-aware reports and support abstention when fixed templates are invalid (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020; Mitchell et al., 2018; AI, 2023; Mahmood et al., 2021).

Reviewers.

- **Enforce semantic and measurement clarity.** Require explicit non-existence encoding with aligned losses and metrics, and credit released canonicalization rules, evaluation code, and slice reporting when a paper claims human-centric generalization (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors; Jin et al., 2020; Gebru et al., 2018; Mitchell et al., 2018; AI, 2023; International Conference on Machine Learning, 2026).

8. Conclusion

Human-centric vision often treats a fixed skeletal topology as the shared interface for supervision, inference, and evaluation, yet a fixed joint inventory does not cover structural absence and a fixed adjacency can be anatomically wrong, which encourages template completion and hallucinated structure for individuals with limb deficiencies. We argue that human-centric vision requires topological generalization beyond fixed skeletons, treating the skeleton graph as an input-conditioned object to infer rather than a constant schema to fit. An instance-adaptive skeletal-topology paradigm represents node existence, typed landmarks, and valid instance-specific connectivity, so evaluation scores structural correctness and penalizes hallucination under absence. We call on dataset curators, benchmark organizers, model builders, and reviewers to adopt absence-explicit semantics and topology-aware measurement so that claimed generalization reflects real morphology rather than canonical completion.

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A. Existence-aware Annotation Protocol

This appendix specifies a minimal protocol that makes anatomical absence explicit and auditable, rather than overloading unlabeled fields or proxy visibility flags (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors; Jin et al., 2020).

A.1. Label space and decision rules

Per-type existence semantics. For each node type in a published vocabulary, annotate exactly one state:

- **Present-visible:** the entity exists and evidence supports localization.
- **Present-occluded:** the entity exists but evidence is missing due to occlusion or coverage.
- **Non-existent:** the entity does not exist for the depicted person.

Decision rules.

- **Occlusion changes evidence, not anatomy.** If the part plausibly exists but is hidden, label present-occluded.
- **Absence changes anatomy.** If the part does not exist, label non-existent even if nearby edges or textures appear.
- **Do not proxy absence as unlabeled.** Encoding non-existence as “not labeled” weakens penalties for hallucinated structure under standard COCO-style pipelines (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors; Jin et al., 2020).

A.2. Endpoint and functional landmark types

Endpoint types. When limb deficiency is present, include at least one endpoint-type landmark per affected limb, such as a residual-limb endpoint (Ying et al., 2025).

Endpoint guidance.

- **Definition:** the endpoint denotes the terminal point of the residual limb that supports functional reasoning.
- **Existence:** endpoints remain valid targets even when distal canonical joints are non-existent.
- **Occlusion:** if the endpoint is covered, label present-occluded unless the residual limb itself is absent.

A.3. Topology ground truth and canonicalization

Ground-truth edges. Provide an instance-level edge set $\mathcal{E}$ in one of two ways:

- **Rule-derived edges:** publish a type-conditional adjacency rule that removes segments involving non-existent nodes and adds endpoint links.
- **Explicit edges:** provide per-instance typed edges with a published directed or undirected convention.

Canonicalization. Publish a deterministic canonicalization that maps annotations and predictions into a shared evaluation space, including type normalization, duplicate handling, node ordering, and edge encoding (Mitchell et al., 2018).

A.4. Slice metadata

Minimum slices. Record metadata that supports comparable slice reporting across papers (Ying et al., 2025):

- Limb level (for example, above-elbow, below-elbow, above-knee, below-knee).
- Laterality (left, right, bilateral).
- Embodiment state (no prosthesis, prosthesis present).
- Evidence context tags (viewpoint class, clothing coverage class).

A.5. Quality control

Agreement targets. Use double annotation on a subset and report agreement for the three-way existence state. Focus adjudication on the boundary between present-occluded and non-existent, and publish decision examples for frequent disagreements. This aligns with transparency practices that make dataset semantics auditable and reusable (Geburu et al., 2018; Mitchell et al., 2018).

B. COCO Keypoints: Visibility Flags Do Not Encode Existence

This appendix section clarifies why COCO-style keypoint fields do not provide anatomical existence semantics (jin-s13 & COCO-WholeBody contributors; cocodataset & COCO API contributors; Jin et al., 2020).

B.1. What the v flag means in practice

COCO uses v for label availability and visibility. In the common COCO keypoint representation, $v \in \{0, 1, 2\}$ indicates whether a keypoint is labeled and whether it is visible, not whether the anatomical entity exists (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020).

B.2. Three downstream consequences

Consequence 1: target semantics remain ambiguous. Setting $v = 0$ removes a keypoint from supervision, but it does not define a distinct semantic state for non-existence. A fixed-index output therefore mixes “unknown label”, “unobserved”, and “non-existent” behind the same training behavior (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020).

Consequence 2: standard scores can under-penalize hallucination. When non-existence maps to unlabeled fields, hallucinated structure in absent regions can contribute weakly or not at all to standard keypoint scores. Reference evaluation logic can ignore instances with no labeled keypoints, which weakens the feedback loop that the data sufficiency argument relies on (cocodataset & COCO API contributors).

Consequence 3: downstream stages cannot consume v as anatomy. Passing $v = 0$ to downstream modules communicates missing annotation, not anatomical absence. A downstream system that needs topology-consistent semantics therefore requires an explicit non-existent state, not a visibility flag (Ying et al., 2025).

Practical fix. Keep COCO comparability, but add an explicit non-existent label as a first-class field, and score hallucination explicitly rather than relying on unlabeled handling (Ying et al., 2025; cocodataset & COCO API contributors).

C. Metric Definitions and Reference Pseudocode

This appendix defines topology-aware metrics that remain meaningful when node inventories and edges vary across individuals.

C.1. Notation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the ground-truth graph. Let $\hat{\mathcal{G}} = (\hat{\mathcal{V}}, \hat{\mathcal{E}})$ denote the predicted graph after canonicalization. Each ground-truth type τ has an existence state $s(\tau) \in \{\text{present-visible}, \text{present-occluded}, \text{non-existent}\}$. After canonicalization, define a type-level existence score

$$\hat{p}(\tau) = \max_{i: \hat{t}_i = \tau} \hat{p}_i, \quad (4)$$

with $\hat{p}(\tau) = 0$ if no predicted node of type τ exists.

C.2. Node existence metrics

Existence F1 per type. For each type τ , define a binary task “ τ exists” versus “ τ is non-existent”. Compute precision, recall, and F1 by thresholding $\hat{p}(\tau)$. Report per-type F1, macro-average across types, and slice-by-slice results (Ying et al., 2025).

Optional present-state agreement. If present-visible versus present-occluded labels are reliable, report conditional accuracy among present nodes.

C.3. Edge metrics

Edge F1 in a canonical typed space. Canonicalize predicted nodes and edges, then compute edge precision, recall, and F1 over typed pairs. Unless a benchmark states otherwise, evaluate edges only among node types whose ground-truth existence is not non-existent.

Typed graph edit distance. Define edit operations on typed nodes and edges, publish fixed costs, and keep costs constant across experiments.

C.4. Hallucination metrics

Standard COCO-style scoring can under-penalize hallucination when absence collapses into unlabeled handling ([cocodataset & COCO API contributors](#)). Report hallucination directly.

Absent-type false positive rate. Let $\mathcal{A}$ denote the set of types labeled non-existent. For threshold δ , define

$$\text{FPR}_{\text{abs}}(\delta) = \frac{1}{|\mathcal{A}|} \sum_{\tau \in \mathcal{A}} \mathbf{1}[\hat{p}(\tau) \geq \delta]. \quad (5)$$

Absent-region confidence summary. For heatmap-style outputs, report the maximum confidence assigned to each non-existent type. Optionally report the same statistic within a tolerance band around annotated endpoints when available ([Ying et al., 2025](#)).

C.5. Reference pseudocode

Canonicalization.

1. Map predicted type names to the published vocabulary.
2. Merge duplicates deterministically, for example keep the highest $\hat{p}_i$ within each type.
3. Order nodes by type, then by a stable tie-breaker, for example x-coordinate.
4. Encode edges in a canonical form, for example sorted typed pairs for undirected edges.

Scoring.

1. Compute existence metrics per type and per slice.
2. Compute edge metrics in the canonical typed space.
3. Compute hallucination metrics and report threshold sensitivity for $\delta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$.

D. Control Suites and Recommended Reporting Items

This appendix lists minimal controls and reporting items that make topology claims comparable across papers ([Gebru et al., 2018](#); [Mitchell et al., 2018](#); [AI, 2023](#)). These items are recommended for clarity and comparability, rather than required by ICML.

D.1. Control suites

Absence versus occlusion control. Construct matched pairs where visual evidence is similar but semantics differ, for example an amputated limb versus a heavily occluded intact limb matched by clothing and viewpoint. Score whether the method separates non-existent from present-occluded outcomes ([Ying et al., 2025](#)).

Connectivity sensitivity control. Test whether removing a segment changes predictions on intact parts. For fixed-topology baselines, implement masking for non-existent types. For instance-adaptive methods, disable edge adaptation as an ablation. Report edge metrics and hallucination metrics under both settings.

Embodiment and context slices. Report existence, edges, and hallucination over embodiment state, clothing coverage, and viewpoint class ([Ying et al., 2025](#)).

Table 2. Recommended reporting items for skeletal-topology-aware evaluation (Geburu et al., 2018; Mitchell et al., 2018; AI, 2023).

Item	What to report
Semantics	Definitions for present-visible, present-occluded, non-existent, plus boundary examples (jin-s13 & COCO-WholeBody contributors; Jin et al., 2020).
Type vocabulary	Node types, including endpoint types, plus mapping rules from outputs to types (Ying et al., 2025).
Canonicalization	Node ordering, duplicate handling, edge encoding, plus released utilities (Mitchell et al., 2018).
Topology metrics	Existence F1 per type, edge F1, optional typed graph edit distance with fixed costs.
Hallucination	$FPR_{abs}(\delta)$ and absent-region confidence summary, plus threshold sensitivity (cocodataset & COCO API contributors).
Slices	Limb level, laterality, embodiment state, plus at least one evidence-context slice (Ying et al., 2025).
Controls	Absence versus occlusion control, connectivity sensitivity control, plus published protocols.
Downstream proxy	At least one proxy that depends on topology, not only coordinate error.

D.2. Reporting table

E. Blueprint Extensions and Ablations

This appendix collects implementation-facing items that otherwise interrupt the main narrative.

E.1. Minimal outputs

Outputs per instance.

- Nodes $\hat{\mathcal{V}} = \{(\hat{t}_i, \hat{p}_i, \hat{\ell}_i)\}_{i=1}^{\hat{K}}$, with type $\hat{t}_i$, existence $\hat{p}_i$, and support $\hat{\ell}_i$.
- Edges $\hat{\mathcal{E}} \subseteq \hat{\mathcal{V}} \times \hat{\mathcal{V}}$, or a routing rule equivalent to an instance-specific adjacency.

E.2. Minimal objective components

Training losses.

- Existence loss that treats non-existent as a distinct target (Ying et al., 2025).
- Localization loss on present nodes only.
- Edge loss when ground-truth edges are available, or a routing loss aligned with canonicalized edges.

E.3. Recommended ablations

Ablation set.

- Fixed skeletal topology baseline.
- Fixed topology plus masking.
- Fixed node set plus existence head, no edge adaptation, no type extension.
- Fixed node set plus existence head, plus limited type extension, no edge adaptation.
- Full instance-adaptive outputs with type extension and input-conditioned connectivity.

F. Mechanism Chain and Failure Taxonomy

F.1. Mechanism chain as test targets

A minimal chain.

schema mismatch $\rightarrow$ undefined targets $\rightarrow$ routing mismatch $\rightarrow$ metric blind spots $\rightarrow$ system-level distortion. (6)

Existence F1 targets undefined targets, edge F1 targets routing mismatch, and hallucination metrics target metric blind spots (Ying et al., 2025; cocodataset & COCO API contributors).

F.2. Qualitative evidence guidance

Failure types and what detects them.

- **Absent-node hallucination:** predicted present nodes for non-existent types, detected by FPR_{abs} in (5).
- **Connectivity distortion under masking:** topology changes on intact parts under masking, detected by the connectivity sensitivity control plus edge F1.
- **Endpoint omission:** missing residual endpoint types, detected by endpoint existence F1 and slice reporting (Ying et al., 2025).
- **Template-forced completion downstream:** forced canonical body when topology mismatches, documented via mismatch policies and abstention behavior (Mahmood et al., 2021).

Qualitative figure template. Group examples by failure type. For each example, show (i) predicted existence states, (ii) predicted edges after canonicalization, and (iii) the corresponding topology and hallucination scores.

G. LLM Use Disclosure

During manuscript preparation, the authors use ChatGPT (access date: January 2026) to assist with English copy-editing and clarity improvements. The tool is not used to generate scientific claims, experimental results, or references. All statements and citations are written and verified by the authors, who take full responsibility for the content. No confidential or unpublished data is provided to the tool.